

Agent-swarm physics: hypothesis compendium

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One RevTeX page per hypothesis follows; click a row to jump to it. Ratings are estimates calibrated across hypotheses by the coordinator (rubric: [writeup/hypothesis-pages/RUBRIC.md](#)). **Completion:** estimated progress of the research direction (idea 5%, round 1 done 35–50%, holdout run 55–65%, robustness and causal designs 70–85%, settled 90–100%). 5% 25% 50% 75% 100% **Faithfulness** (0–5, scoped): how well the model holds for what it claims (3 = descriptive or primary holdout passed; 4 = supported; 5 = faithful and mechanistic). **Usefulness** (0–5, unscoped): what an operator or alignment researcher can do with it, true or not (1 = vocabulary only; 2 = a diagnostic; 3 = a validated monitor or design rule; 4 = a steering lever; 5 = a transferable control knob). 0

1 2 2.5 3 4 5

ID	Hypothesis	Complete	Faithful	Useful	p.
H01	Emergent superagents exist Rooms are more like-minded than random groups but missed both locked sizes; order looks like room fields	25%	2	2	3
H02	Inferred couplings reflect real influence Activity-timing Ising couplings are null-level and missed the installed #45 leader; only shared hulls remain	60%	0.5	1.5	4
H03	Swarm activity is self-exciting, and its criticality is set by the go... Subcritical everywhere (n about 0.2-0.4), no coupling-mode effect; only fast cross-agent triggering beats nulls	40%	1.5	2	5
H04	External forcing reshapes the response kernel, reversibly Nudges act only on their target, after minutes of delay; the hours-reversal holdout was falsified in reverse	60%	1	2.5	6
H05	Cutting the cross-room channel lowers entropy production, and rooms b... Rooms cut chat coupling (holdout DiD passed; verdict inconclusive); work isn't room-bound; EP unmeasurable	60%	2.5	2.5	7
H06	Free weeks show neutral cooperative dynamics with a cooperator core (draft) specified (shortlist S6 / HH42; approved by Vivian 2026-10-03; started 2026-10-04). Exploratory round 1 starting; pre...	—	—	—	8
H07	The RPG forks diverge from a common ancestor at a measurable rate Forks diverge gradually in code, in team-chosen jumps in content; copying dominates; no leakage; n = 2	40%	2	1.5	9
H08	Context is the coupling Parked, untested: is context inclusion the coupling? H04, H15, H18 hint yes; decisive tests need prompts	5%	0	1.5	10
H09	Agent swarms have an effective thermodynamics Thermo framing adds little; idle agents are timer-gated (only @-mentions get through); memory is a set point	25%	1.5	2	11
H10	Goals are Legendre pushes: a swarm's response to an assigned goal is ... (draft) running (exploratory round 1). Predictions written 2026-10-03 before any real-data run (below).	—	—	—	12
H11	Division of labor vs herding is the sign of a Potts coupling, set by ... Agents herd onto shared projects in 11/14 weeks whatever the goal; spread appears only as fixed ownership.	40%	1.5	2	13
H12	Grouphink is dimensional collapse: few collective modes, shrinking p... One market mode, no kickoff collapse; low-dimensional weeks are agents looping on their own text.	42%	1.5	2.5	14

ID	Hypothesis	Complete	Faithful	Useful	p.
H13	Model families carry their own fields, and couple by family Model families share a stable content field that is writing style; no family homophily; rooms carry coupling.	45%	1.5	2	15
H14	Entropy production of behavior-state sequences: per-family arrows of ... Irreversibility lives in fine tool actions and scaffold steps; no per-family (HH19) or collective (HH67) arrow	45%	1.5	1.5	16
H15	Semantic information through natural scrambles: which information kee... Losing memory costs nothing visible in a day; wiping the context window costs ~45% of output for 10 turns	40%	1.5	2.5	17
H16	Agents get stuck in metastable traps and escape by Kramers kinetics; ... Agent traps age, not Kramers-escape; one directed message at a pause gate helps, more add little.	42%	1.5	2.5	18
H17	Behavior is a Markov state model with a few metastable sets; its mixi... Action-class behavior is non-Markov; slow sets are GUI/shell modes and idling; t2* doesn't flag stuck periods	40%	1	1.5	19
H18	Attention dilutes as 1/k: response to a message falls with the number... Attention dilutes as ~k ^{-0.6} in all 16 periods (not 1/k): per-pair engagement falls with room size	42%	2	2	20
H19	One curve for all periods: per-period loop gains collapse onto a func... No common loop-gain curve: activity and talk channels move oppositely; Hawkes-to-Ising mapping holds.	40%	1	1.5	21
H20	Aging: day-to-day content autocorrelation depends on time since kicko... No aging: swarm content settles within ~4 days of a kickoff, then decorrelates at an age-independent rate	40%	1	1.5	22
H21	The debate week (#12) is a two-sublattice antiferromagnet (draft) running (exploratory round 1, 2026-10-03). Promoted 2026-10-03 by Vivian from HH103.	—	—	—	23
H22	Private, conflicting goals make #51 a spin glass #51 is no spin glass: roles pin positions, couplings are mostly positive, rivals co-move, no collective states	40%	1	1.5	24
H23	Distilling the village into a fine-tuned leader copies its vocabulary... Distilled leader kept a trace of its corpus style but none of its plans; its corpus was Kimi talking to itself	40%	1.5	1.5	25
H24	In the forecast week (#21), switching from independent drafts to comp... Forecast week: the coupling switch was real but early; alignment didn't jump and herding was an artifact	40%	1	1.5	26